Huawei Access Node QoS/Congestion Event Troubleshooting and Resolution Standard Operating Procedure

Node: JAY-ACC-68 | **Vendor:** Huawei | **Severity:** Major | **Alarm Code:** ALARM_1671

Description

The QoS/Congestion Event failure domain, specifically identified by alarm code ALARM_1671, on a Huawei Access Node (e.g., Node ID: JAY-ACC-68 at Jayanagar Metro Site), occurs due to transient or sustained periods of microbursts, interface tail drops, or QoS queue starvation. This phenomenon can be attributed to the intricacies of packet scheduling algorithms, buffer management, and the complex interplay between various traffic classes. On Huawei equipment, the proprietary VRP (Versatile Routing Platform) operating system orchestrates these processes. When the network experiences a surge in traffic that exceeds the allocated bandwidth for a particular class, or when there's a misconfiguration in the QoS policy, it can lead to queue overflows and packet drops, triggering the ALARM_1671. Understanding the root causes requires a deep dive into the specifics of Huawei's QoS implementation, including the Comprehensive QoS (ComQoS) model, which manages traffic through a hierarchical policy framework. The failure is critical as it impacts user experience, potentially leading to timeouts, loss of connectivity, and decreased network throughput.

Pre-requisite Checks

- Verify the physical integrity of the site and equipment at Jayanagar Metro Site.
- Check the status of redundant links to ensure no single point of failure.
- Confirm that all necessary tools and Huawei-specific diagnostic software are available.

Potential Root Causes

Cause	Probability
Misconfigured QoS Policy	High
Insufficient Bandwidth Allocation	Medium
Hardware Failure (e.g., Line Card Issue)	Low

Diagnostic Commands & Expected Outputs

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Command: display qos interface, Expected Output: QoS interface configuration and statistics, Description: This command is used to display the QoS configuration and statistics of the interfaces. It helps in identifying any misconfigurations or issues related to the QoS policy applied on the interfaces.

Command: display traffic-policy applied-record, Expected Output: Applied traffic policy records, Description: This command displays the
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records of applied traffic policies, which can help in identifying if there are any issues with the policy application or if the policies are correctly applied on the interfaces.

Command: display interface brief, Expected Output: Brief interface status and statistics, Description: This command provides a brief overview of the interface status and statistics, which can help in identifying any interface-level issues that might be contributing to the QoS/Congestion Event.
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Resolution Steps

Step 1: Step: Adjust QoS Policy, Command: qos policy class priority , Description: Adjust the QoS policy to ensure proper traffic classification and prioritization. This involves modifying the existing policy to allocate appropriate bandwidth and priority to critical traffic classes.

Step 2: Step: Allocate Sufficient Bandwidth, Command: interface bandwidth , Description: Allocate sufficient bandwidth to the interface to handle the traffic volume. This involves adjusting the interface bandwidth to match the expected traffic load, preventing congestion and queue overflows.

Step 3: Step: Apply Traffic Policy, Command: traffic-policy inbound, Description: Apply the adjusted traffic policy to the relevant interface in the inbound direction to ensure that the policy takes effect and traffic is managed according to the new configuration.

Post-Check Verification

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Command: display qos interface, Expected Output: Updated QoS interface configuration and statistics reflecting the changes made, Description: Verify that the QoS policy changes have taken effect and that the interface statistics reflect the expected behavior post-configuration adjustments.

Command: display traffic-policy applied-record, Expected Output: Updated applied traffic policy records, Description: Confirm that the updated traffic policy is correctly applied and reflected in the system records.
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Roll-back Procedures

- Step: Revert QoS Policy, Command: undo qos policy , Description: Revert the QoS policy to its previous state in case the adjustments made do not resolve the issue or introduce new problems. This ensures that the system returns to a known state.

- Step: Restore Original Bandwidth, Command: undo interface bandwidth, Description: Restore the original bandwidth allocation to the interface to prevent any unintended bandwidth-related issues that might have been introduced during the troubleshooting process.